

Task-1

1. Browser: HTTP Version 1.1
Server: HTTP Version 1.1
2. Accept-Language: en-IN,en-GB;q=0.9,en;q=0.8\r\n
3. Server: 10.70.48.108
Destination: 128.119.245.12
4. 200 OK
5. Date: Tue, 28 Jan 2025 05:49:47 GMT\r\n
6. File Data: 218 bytes
7. No header found

The image shows a Wireshark packet capture of an HTTP transaction. The top pane displays a list of packets, with packet 33 selected. The middle pane shows the details of the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
33	10.422550	10.70.48.108	128.119.245.12	HTTP	472	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
42	10.639176	128.119.245.12	10.70.48.108	HTTP	540	HTTP/1.1 200 OK (text/html)
66	10.870248	10.70.48.108	128.119.245.12	HTTP	312	GET /apple-touch-icon-precomposed.png HTTP/1.1
83	11.080516	128.119.245.12	10.70.48.108	HTTP	560	HTTP/1.1 404 Not Found (text/html)
85	11.083816	10.70.48.108	128.119.245.12	HTTP	300	GET /apple-touch-icon.png HTTP/1.1
86	11.366687	128.119.245.12	10.70.48.108	HTTP	547	HTTP/1.1 404 Not Found (text/html)

Packet 33 Details:

- Frame 33: 472 bytes on wire (3776 bits), 472 bytes captured (3776 bits) on interface en0, Ethernet II, Src: Apple_4d:fa:19 (1c:57:dc:4d:fa:19), Dst: 12:01:00:00:01:00 (12:01:00:00:01:00)
- Internet Protocol Version 4, Src: 10.70.48.108, Dst: 128.119.245.12
- Transmission Control Protocol, Src Port: 58147, Dst Port: 80, Seq: 1, Ack: 1, Len: 418
- Hypertext Transfer Protocol
 - GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
 - Host: gaia.cs.umass.edu\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
 - User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/17.5 Safari/605.1.15
 - Accept-Language: en-IN,en-GB;q=0.9,en;q=0.8\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Connection: keep-alive\r\n
 - \r\n

Raw Data (Hex):

```
0000 12 01 00 00 01 00 1c 57 dc 4d fa 19 08 00 45 00 .....W.M....E.
0010 01 ca 00 00 40 00 40 06 88 f8 0a 46 30 6c 80 77 ...@...F0L.w
0020 f5 0c e3 23 00 50 fd f8 e6 c2 a6 c6 97 05 50 18 ...#P.....P.
0030 10 00 7f 5f 00 00 47 45 54 20 2f 77 69 72 65 73 ...GET/wires
0040 68 61 72 6b 2d 6c 61 62 73 2f 48 54 54 50 2d 77 harK-lab s/HTTP-w
0050 69 72 65 73 68 61 72 6b 2d 66 69 6c 65 31 2e 68 ireshark -file1.h
0060 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f tml HTTP /1.1 .Ho
0070 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 73 st: gaia .cs.umas
0080 73 2e 65 64 75 0d 0a 55 70 67 72 61 64 65 2d 49 s.edu .U pgrade-I
0090 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73 74 73 nsecure- Requests
00a0 3a 20 31 0d 0a 41 63 63 65 70 74 3a 20 74 65 78 : 1·Acc ept: tex
00b0 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 74 69 t/html,a pplicati
00c0 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70 70 on/xhtmll +xml,app
00d0 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 3d 30 lication /xml;q=0
00e0 2e 39 2c 2a 2f 2a 3b 71 3d 30 2e 38 0d 0a 55 73 .9,*/*;q =0.8·Us
00f0 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c er-Agent : Mozill
0100 61 2f 35 2e 30 20 28 4d 61 63 69 6e 74 6f 73 68 a/5.0 (Macintosh
0110 3b 20 49 6e 74 65 6c 20 4d 61 63 20 4f 53 20 59 ; Intel Mac OS X
0120 20 31 30 5f 31 35 5f 37 29 20 41 70 70 6c 65 57 10.15.7 ) AppleW
0130 65 62 4b 69 74 2f 36 30 35 2e 31 2e 31 35 20 28 ebKit/60 5.1.15 (
0140 4b 48 54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63 6b KHTML, l ike Geck
0150 6f 29 20 56 65 72 73 69 6f 6e 2f 31 37 2e 35 20 o) Versi on/17.5
0160 53 61 66 61 72 69 2f 36 30 35 2e 31 2e 31 35 0d Safari/6 05.1.15·
0170 0a 41 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 65 ·Accept- Language
0180 3a 20 65 6e 2d 49 4e 2c 65 6e 2d 47 42 3b 71 3d en-IN, en-GB;q=
0190 30 2e 39 2c 65 6e 3b 71 3d 30 2e 38 0d 0a 41 63 0.9,en;q =0.8·Ac
01a0 63 65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 cept-Enc oding: g
01b0 7a 69 70 2c 20 64 65 66 6c 61 74 65 0d 0a 43 6f zip, def late: Co
```

Status Bar: *http1* is not a valid protocol or protocol field. Packets: 88 · Displayed: 6 (6.8%) · Dropped: 0 (0.0%) Profile: Default

Task-2

1. No. We do not see an IF-MODIFIED-SINCE as the file is requested for 1st time.
2. Yes, it returned the contents of the file. The server response is 200 OK which means that all the file contents were returned explicitly.
3. Yes, we see an IF-MODIFIED-SINCE in the 2nd HTTP GET request, as the website was refreshed. A date and time follows the phrase to indicate the file has not changed from that time.
4. The status code and phrase returned are '304 NOT MODIFIED' in response to the second HTTP GET request. No, the contents of the file were not explicitly returned this time as the file was not modified since the timestamp.

Wireshark packet capture showing HTTP traffic. The packet list shows a GET request for a file, a 200 OK response, and a subsequent GET request that receives a 304 Not Modified response. The packet details for the 304 response show the 'Not Modified' status and the 'Date' and 'ETag' headers.

No.	Time	Source	Destination	Protocol	Length	Info
165	3.062652	10.70.51.84	142.250.183.227	OCSP	543	Request
167	3.111754	142.250.183.227	10.70.51.84	OCSP	756	Response
214	3.997597	10.70.51.84	128.119.245.12	HTTP	492	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
216	4.206624	128.119.245.12	10.70.51.84	HTTP	784	HTTP/1.1 200 OK (text/html)
249	6.548950	10.70.51.84	128.119.245.12	HTTP	578	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
250	6.756681	128.119.245.12	10.70.51.84	HTTP	293	HTTP/1.1 304 Not Modified

Frame 250: 293 bytes on wire (2344 bits), 293 bytes captured (2344 bits) on interface...
Ethernet II, Src: HewlettPacka_ac:4d:00 (e4:de:40:ac:4d:00), Dst: AzureWaveTec_e4:90...
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.70.51.84
Transmission Control Protocol, Src Port: 80, Dst Port: 51485, Seq: 731, Ack: 963, Len: 293
[Stream index: 7]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 239]
Sequence Number: 731 (relative sequence number)
Sequence Number (raw): 569702253
[Next Sequence Number: 970 (relative sequence number)]
Acknowledgment Number: 963 (relative ack number)
Acknowledgment number (raw): 3385241992
0101 = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)

0000 34 f2 24 e4 90 2d e4 de 40 ac 4d 00 08 00 45 00 40\$--- @M..E
0010 01 17 ce 5b 40 00 30 06 c8 67 80 77 f5 0c 0a 45 ...[@. g w..F
0020 33 54 00 50 c9 1d 21 f4 f7 6d c9 c6 b1 88 50 18 3T P. l. m. . P
0030 00 f5 00 f5 00 00 48 54 50 2f 31 2e 31 20 33 00 ... HT TP/1.1 3
0040 30 34 20 4e 6f 74 20 4d 6f 64 69 66 69 65 64 0d 04 Not M odified
0050 0a 44 61 74 65 3a 20 54 75 65 2c 20 32 38 20 4a Date: Tue, 28 J
0060 61 6e 20 32 30 32 35 20 30 36 3a 30 37 3a 31 39 an 2025 06:07:19
0070 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 20 41 70 GMT Se rver: Ap
0080 61 63 68 65 2f 32 2e 34 2e 36 20 28 43 65 6e 74 ache/2.4 .6 (Cent
0090 4f 53 29 20 4f 70 65 6e 53 53 4c 2f 31 2e 30 2e OS) Open SSL/1.0.
00a0 32 6b 2d 66 69 70 73 20 50 48 50 2f 37 2e 34 2e 2k-fips PHP/7.4.
00b0 33 33 20 6d 6f 64 5f 70 65 72 6c 2f 32 2e 30 2e 33 mod_p erl/2.0.
00c0 31 31 20 50 65 72 6c 2f 76 35 2e 31 36 2e 33 0d 11 Per1/ v5.16.3
00d0 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 Connect ion: Kee
00e0 70 2d 41 6c 69 76 65 0d 0a 4b 65 65 70 2d 41 6c p-Alive: Keep-Al
00f0 69 76 65 3a 20 74 69 6d 65 6f 75 74 3d 35 2c 20 ive: tim eout=5,
0100 6d 61 78 3d 39 3d 0d 0a 45 54 61 67 3a 20 22 31 max=99 ETag: "1
0110 37 33 2d 36 32 63 62 65 30 30 35 62 36 36 34 37 73-62cbe 005b6647

Packets: 275 · Displayed: 6 (2.2%) · Dropped: 0 (0.0%) Profile: Default

Task-3

1. The browser sent 1 GET Request message. The packet number in the trace containing the Bill of Rights is #27.
2. The trace that contains the status code and phrase associated with the response to the HTTP GET request is #26.
3. The status code and phrase is 200 OK.
4. 4 data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

The image shows a Wireshark network traffic capture. The top pane displays a list of packets. Packet 24 is a GET request for /wireshark-labs/HTTP-wireshark-file3.html. Packet 26 is the HTTP response with status 200 OK. Packet 27 is the first of four TCP segments carrying the response body.

No.	Time	Source	Destination	Protocol	Length	Info
24	10.440681	10.70.48.108	128.119.245.12	HTTP	472	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
29	10.748144	128.119.245.12	10.70.48.108	HTTP	769	HTTP/1.1 200 OK (text/html)

Packet 26 details:

- Frame 26: 769 bytes on wire (6152 bits), 769 bytes captured
- Ethernet II, Src: HewlettPacka_ac:4d:00 (e4:de:40:ac:4d:00), Dst: 10.70.48.108
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.70.48.108
- Transmission Control Protocol, Src Port: 80, Dst Port: 53438
- [4 Reassembled TCP Segments (4861 bytes): #26(1382), #27(1382), #28(1382), #29(715)]
- [Frame: 26, payload: 0-1381 (1382 bytes)]
- [Frame: 27, payload: 1382-2763 (1382 bytes)]
- [Frame: 28, payload: 2764-4145 (1382 bytes)]
- [Frame: 29, payload: 4146-4860 (715 bytes)]
- [Segment count: 4]
- [Reassembled TCP length: 4861]
- [Reassembled TCP Data [...]: 485454502f312e3120323030204f4b0d0a446174653a205475652c203238204a616e20323032352031343a35353a31372020474d540d0a5365727065727a3a204170616368652f322e342e36202843656e744f5329204f70656553534c2f312e302e326b2d66697073205048502f372e342e3333206d6f645f7065726c2f322e302e3131205065726c2f76352e31362e330d0a4c6173742d4d6f6469666965643a202030363a35393a303120474d540d0a455461673a2022313139342d36326362656261346531366134220d0a4163636570742d52616e6765733a2062797465730d0a436f6e74656e742d4c656e6774683a20343530300d0a4b6565702d416c697675743d352c206d616f6e6e656374696f6f6e6e65637a696f6e6e6976650d0a437970653a20746578636861727365743d0a3c68746d6c3e3c686561643e200a3c7469746c653e486973746f726963616c20446f63756d656e74733a5448452042494c4c204f46205249474854533c2f7469746c653e3c2f63626f6479206267636f6c6f723d2223666666666622206c696e6b3d222333303030302220766c

Packet 27 details:

- Frame 27: 1382 bytes on wire (11056 bits), 1382 bytes captured
- Ethernet II, Src: HewlettPacka_ac:4d:00 (e4:de:40:ac:4d:00), Dst: 10.70.48.108
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.70.48.108
- Transmission Control Protocol, Src Port: 80, Dst Port: 53438
- [4 Reassembled TCP Segments (4861 bytes): #26(1382), #27(1382), #28(1382), #29(715)]
- [Frame: 27, payload: 1382-2763 (1382 bytes)]
- [Frame: 28, payload: 2764-4145 (1382 bytes)]
- [Frame: 29, payload: 4146-4860 (715 bytes)]
- [Segment count: 4]
- [Reassembled TCP length: 4861]
- [Reassembled TCP Data [...]: 485454502f312e3120323030204f4b0d0a446174653a205475652c203238204a616e20323032352031343a35353a31372020474d540d0a5365727065727a3a204170616368652f322e342e36202843656e744f5329204f70656553534c2f312e302e326b2d66697073205048502f372e342e3333206d6f645f7065726c2f322e302e3131205065726c2f76352e31362e330d0a4c6173742d4d6f6469666965643a202030363a35393a303120474d540d0a455461673a2022313139342d36326362656261346531366134220d0a4163636570742d52616e6765733a2062797465730d0a436f6e74656e742d4c656e6774683a20343530300d0a4b6565702d416c697675743d352c206d616f6e6e656374696f6f6e6e65637a696f6f6e6e6976650d0a437970653a20746578636861727365743d0a3c68746d6c3e3c686561643e200a3c7469746c653e486973746f726963616c20446f63756d656e74733a5448452042494c4c204f46205249474854533c2f7469746c653e3c2f63626f6479206267636f6c6f723d2223666666666622206c696e6b3d222333303030302220766c

Packet 28 details:

- Frame 28: 1382 bytes on wire (11056 bits), 1382 bytes captured
- Ethernet II, Src: HewlettPacka_ac:4d:00 (e4:de:40:ac:4d:00), Dst: 10.70.48.108
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.70.48.108
- Transmission Control Protocol, Src Port: 80, Dst Port: 53438
- [4 Reassembled TCP Segments (4861 bytes): #26(1382), #27(1382), #28(1382), #29(715)]
- [Frame: 28, payload: 2764-4145 (1382 bytes)]
- [Frame: 29, payload: 4146-4860 (715 bytes)]
- [Segment count: 4]
- [Reassembled TCP length: 4861]
- [Reassembled TCP Data [...]: 485454502f312e3120323030204f4b0d0a446174653a205475652c203238204a616e20323032352031343a35353a31372020474d540d0a5365727065727a3a204170616368652f322e342e36202843656e744f5329204f70656553534c2f312e302e326b2d66697073205048502f372e342e3333206d6f645f7065726c2f322e302e3131205065726c2f76352e31362e330d0a4c6173742d4d6f6469666965643a202030363a35393a303120474d540d0a455461673a2022313139342d36326362656261346531366134220d0a4163636570742d52616e6765733a2062797465730d0a436f6e74656e742d4c656e6774683a20343530300d0a4b6565702d416c697675743d352c206d616f6e6e656374696f6f6e6e65637a696f6f6e6e6976650d0a437970653a20746578636861727365743d0a3c68746d6c3e3c686561643e200a3c7469746c653e486973746f726963616c20446f63756d656e74733a5448452042494c4c204f46205249474854533c2f7469746c653e3c2f63626f6479206267636f6c6f723d2223666666666622206c696e6b3d222333303030302220766c

Packet 29 details:

- Frame 29: 715 bytes on wire (5720 bits), 715 bytes captured
- Ethernet II, Src: HewlettPacka_ac:4d:00 (e4:de:40:ac:4d:00), Dst: 10.70.48.108
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.70.48.108
- Transmission Control Protocol, Src Port: 80, Dst Port: 53438
- [4 Reassembled TCP Segments (4861 bytes): #26(1382), #27(1382), #28(1382), #29(715)]
- [Frame: 29, payload: 4146-4860 (715 bytes)]
- [Segment count: 4]
- [Reassembled TCP length: 4861]
- [Reassembled TCP Data [...]: 485454502f312e3120323030204f4b0d0a446174653a205475652c203238204a616e20323032352031343a35353a31372020474d540d0a5365727065727a3a204170616368652f322e342e36202843656e744f5329204f70656553534c2f312e302e326b2d66697073205048502f372e342e3333206d6f645f7065726c2f322e302e3131205065726c2f76352e31362e330d0a4c6173742d4d6f6469666965643a202030363a35393a303120474d540d0a455461673a2022313139342d36326362656261346531366134220d0a4163636570742d52616e6765733a2062797465730d0a436f6e74656e742d4c656e6774683a20343530300d0a4b6565702d416c697675743d352c206d616f6e6e656374696f6f6e6e65637a696f6f6e6e6976650d0a437970653a20746578636861727365743d0a3c68746d6c3e3c686561643e200a3c7469746c653e486973746f726963616c20446f63756d656e74733a5448452042494c4c204f46205249474854533c2f7469746c653e3c2f63626f6479206267636f6c6f723d2223666666666622206c696e6b3d222333303030302220766c

Task-4

1. 3 HTTP GET Requests messages were sent by the browser. The Internet Addresses are as follows: 128.119.245.12 and 178.79.137.64
2. The browser downloaded the images serially. This can be said as the timestamps are different.

The image shows a Wireshark network traffic capture. The top pane displays a list of captured packets, with the first three being HTTP GET requests. The bottom pane shows a detailed view of the selected packet (No. 41), including the Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol layers. The packet data is displayed in hexadecimal and ASCII format.

No.	Time	Source	Destination	Protocol	Length	Info
41	12.640237	10.70.48.108	128.119.245.12	HTTP	472	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
44	12.898441	128.119.245.12	10.70.48.108	HTTP	1355	HTTP/1.1 200 OK (text/html)
48	12.925564	10.70.48.108	128.119.245.12	HTTP	550	GET /pearson.png HTTP/1.1
60	13.078340	10.70.48.108	178.79.137.64	HTTP	517	GET /8E_cover_small.jpg HTTP/1.1
63	13.137923	128.119.245.12	10.70.48.108	HTTP	901	HTTP/1.1 200 OK (PNG)
66	13.220159	178.79.137.64	10.70.48.108	HTTP	225	HTTP/1.1 301 Moved Permanently

> Frame 41: 472 bytes on wire (3776 bits), 472 bytes captured (3776 bits) on interface en0, id 0
> Ethernet II, Src: Apple_4d:fa:19 (1c:57:dc:4d:fa:19), Dst: 12:01:00:00:01:00 (12:01:00:00:01:00)
> Internet Protocol Version 4, Src: 10.70.48.108, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 53467, Dst Port: 80, Seq: 1, Ack: 1, Len: 418
> Hypertext Transfer Protocol

0000 12 01 00 00 01 00 1c 57 dc 4d fa 19 08 00 45 00W.M....E.
0010 01 ca 00 00 40 00 00 06 88 f8 0a 46 30 6c 80 77@...F0L.w
0020 f5 0c d0 db 00 50 ee 71 c7 2e 56 3c f7 f6 50 18P.q..Vc..P.
0030 10 00 b0 58 00 00 47 45 54 20 2f 77 69 72 65 73 ...X...GE T /wires
0040 68 61 72 6b 2d 6c 61 62 73 2f 48 54 54 50 2d 77 hark-lab s/HTTP-w
0050 69 72 65 73 68 61 72 6b 2d 66 69 6c 65 34 2e 68 iredshark -file4,h
0060 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48 6f tml HTTP /1.1 Ho
0070 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 73 st: gaia .cs.umas
0080 73 2e 65 64 75 0d 0a 55 70 67 72 61 64 65 2d 49 s.edu U pgrade-I
0090 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73 74 73 nsecure- Requests
00a0 3a 20 31 0d 0a 41 63 63 65 70 74 3a 20 74 65 78 : 1 Acc ept: tex
00b0 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 74 69 t/html,a pplicati
00c0 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70 70 on/xhtml+xml,app
00d0 6c 69 63 61 74 69 6f 6e 2f 78 6d 6c 3b 71 3d 30 lication /xml;q=0
00e0 2e 39 2c 2a 2f 2a 3b 71 3d 30 2e 38 0d 0a 55 73 .9,*/*;q=0.8 Us
00f0 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c er-Agent : Mozill
0100 61 2f 35 2e 30 20 28 4d 61 63 69 6e 74 6f 73 68 a/5.0 (M acintosh
0110 3b 20 49 6e 74 65 6c 20 4d 61 63 20 4f 53 20 58 ; Intel Mac OS X
0120 20 31 30 5f 31 35 5f 37 29 20 41 70 70 6c 65 57 10 15.7) AppleW
0130 65 62 4b 69 74 2f 36 30 35 2e 31 2e 31 35 20 28 ebKit/60 5.1.15 (K
0140 4b 48 54 4d 4c 2c 20 6c 69 6b 65 20 47 65 63 6b HTML, l ike Geck
0150 6f 29 20 56 65 72 73 69 6f 6e 2f 31 37 2e 35 20 o) Versi on/17.5
0160 53 61 66 61 72 69 2f 36 30 35 2e 31 2e 31 35 0d Safari/6 05.1.15
0170 0a 41 63 63 65 70 74 2d 4c 61 6e 67 75 61 67 65 Accept- Language
0180 3a 20 65 6e 2d 49 4e 2c 65 6e 2d 47 42 3b 71 3d : en-IN, en-GB;q=
0190 30 2e 39 2c 65 6e 3b 71 3d 30 2e 38 0d 0a 41 63 0.9,en;q =0.8 Ac
01a0 63 65 70 74 2d 45 6e 63 6f 64 69 6e 67 3a 20 67 cept=Enc oding: g
01b0 7a 69 70 2c 20 64 65 66 6c 61 74 65 0d 0a 43 6f zip, def late Co
01c0 6e 6e 65 63 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 nnection : keep-a
01d0 6c 69 76 65 0d 0a 0d 0a 6c 20 6b 65 65 70 2d 61 live....

Task-5

1. Before entering the details i.e. the username and password, the server's response was "401 Unauthorized".
2. The new field that was included in the HTTP GET message for the second time was the Line-based text data. It revealed the text in the website after logging in.

The image shows a Wireshark packet capture window. The top pane displays a list of packets. Packet 123 is highlighted, showing an HTTP 401 Unauthorized response. Packet 173 is also highlighted, showing an HTTP GET request with line-based text data.

No.	Time	Source	Destination	Protocol	Length	Info
120	35.250664	10.70.48.108	128.119.245.12	HTTP	488	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
123	35.457529	128.119.245.12	10.70.48.108	HTTP	771	HTTP/1.1 401 Unauthorized (text/html)
171	47.891427	10.70.48.108	128.119.245.12	HTTP	547	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
173	48.123023	128.119.245.12	10.70.48.108	HTTP	544	HTTP/1.1 200 OK (text/html)

Packet 173 details:

- > Frame 173: 544 bytes on wire (4352 bits), 544 bytes captured (4352 bits) on interface en0, id 0
- > Ethernet II, Src: HewlettPacka_ac:4d:00 (e4:de:40:ac:4d:00), Dst: Apple_4d:fa:19 (1c:57:dc:4d:fa:19)
- > Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.70.48.108
- > Transmission Control Protocol, Src Port: 80, Dst Port: 53486, Seq: 1, Ack: 494, Len: 490
- > Hypertext Transfer Protocol
- > Line-based text data: text/html (6 lines)

Packet 173 hex data:

```
0000 1c 57 dc 4d fa 19 e4 de 40 ac 4d 00 08 00 45 00 .W.M....@M...E-
0010 02 12 4d 7d 40 00 30 06 4b 33 80 77 f5 0c 0a 46 .M}@0.K3w...F
0020 30 6c 00 50 d0 ee b9 13 de bd 44 c0 32 05 50 18 0lP....D.2.P.
0030 00 ed 5f 30 00 00 48 54 54 50 2f 31 2e 31 20 32 ..0..HT TP/1.1 2
0040 30 30 20 4f 4b 0d 0a 44 61 74 65 3a 20 54 75 65 00 OK..D ate: Tue
0050 2c 20 32 38 20 4a 61 6e 20 32 30 32 35 20 31 35 , 28 Jan 2025 15
0060 3a 31 35 3a 35 38 20 47 4d 54 0d 0a 53 65 72 76 ;15:58 GMT. Serv
0070 65 72 3a 20 41 70 61 63 68 65 2f 32 2e 34 2e 36 er: Apac he/2.4.6
0080 20 28 43 65 6e 74 4f 53 29 20 4f 70 65 6e 53 53 (CentOS ) OpenSS
0090 4c 2f 31 2e 30 2e 32 6b 2d 66 69 70 73 20 50 48 L/1.0.2k -fips PH
00a0 50 2f 37 2e 34 2e 33 33 20 6d 6f 64 5f 70 65 72 P/7.4.33 mod_per
00b0 6c 2f 32 2e 30 2e 31 31 20 50 65 72 6c 2f 76 35 L/2.0.11 Perl/v5
00c0 2e 31 36 2e 33 0d 0a 4c 61 73 74 2d 4d 6f 64 69 .16.3..L ast-Modi
00d0 66 69 65 64 3a 20 54 75 65 2c 20 32 38 20 4a 61 fied: Tue, 28 Ja
00e0 6e 20 32 30 32 35 20 30 36 3a 35 39 3a 30 31 20 n 2025 0 6:59:01
00f0 47 4d 54 0d 0a 45 54 61 67 3a 20 22 38 34 2d 36 GMT..ETa g: "84-6
0100 32 63 62 65 62 61 34 65 36 30 64 63 22 0d 0a 41 2cbeba4e 60dc"...A
0110 63 63 65 70 74 2d 52 61 6e 67 65 73 3a 20 62 79 ccept-Ra nges: by
0120 74 65 73 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 65 6e tes..Con tent-Len
0130 67 74 68 3a 20 31 33 32 0d 0a 4b 65 65 70 2d 41 gth: 132 ..Keep-A
0140 6c 69 76 65 3a 20 74 69 6d 65 6f 75 74 3d 35 2c live: ti meout=5,
0150 20 6d 61 78 3d 31 30 30 0d 0a 43 6f 6e 6e 65 63 max=100 ..Connec
0160 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 6c 69 76 65 tion: Ke ep-Alive
0170 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 ..Conten t-Type:
0180 74 65 78 74 2f 68 74 6d 6c 3b 20 63 68 61 72 73 text/htm l; chars
0190 65 74 3d 55 54 46 2d 38 0d 0a 0d 0a 0a 3c 68 74 et=UTF-8 .....<ht
01a0 6d 6c 3e 0a 0a 54 68 69 73 20 70 61 67 65 20 69 ml>..Thi s page i
01b0 73 20 70 61 73 73 77 6f 72 64 20 70 72 6f 74 65 s passwo rd prote
01c0 63 74 65 64 21 20 20 49 66 20 79 6f 75 27 72 65 cted! I f you're
01d0 20 73 65 65 69 6e 67 20 74 68 69 73 2c 20 79 6f seeing this, yo
01e0 75 27 76 65 20 64 6f 77 6e 6c 6f 61 64 65 64 20 u've dow nloaded
```